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Minimum wage increases and agricultural employment of locals and guest workers

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Abstract

We estimate the impact of the minimum wage on US farm worker employment by exploiting state-level variation in recent minimum wage increases. A $1.00 increase in the minimum wage decreases the likelihood of employment for an individual farm worker by around one percentage point, with the effect concentrated among native and naturalized US citizens. However, we find no evidence of a relationship between minimum wage and either the number of farm workers or guest workers. The difference between these results may be due to lower international migration to high minimum wage states. Citizenship and immigration status can influence the impact of labor market regulations on the farm workforce.

KEYWORDS

agricultural employment, farm labor, minimum wage

JEL CLASSIFICATION

J43: agricultural labor markets, Q18: agricultural policy; food policy

1 | INTRODUCTION

Labor costs are a major expense for many farms. For example, for specialty crop production, labor costs are on average 40% of US farm expenses (USDA Economic Research Service). Increasing domestic labor regulation, tight nonfarm labor markets, and ongoing political gridlock on immigration policy influence domestic agricultural labor shortages (Charlton et al., 2019b; Hertz & Zahniser, 2013; Ifft & Grout, 2017; Richards, 2018; Roka & Guan, 2018; Taylor & Charlton, 2019; Zahniser et al., 2019). These upward pressures on farm wages have long-term implications for farm structure as well as domestic and global patterns of food production and consumption. Minimum

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wage hikes, which have recently occurred in several major agricultural states (Ifft & Grout, 2017), have direct impacts on agricultural labor, as about 40% of farm workers are paid at or near the minimum wage (Kandilov & Kandilov, 2020). However, the impact of rising wages on agricultural labor markets is uncertain due to the diversity of the US agricultural labor force. Findings are mixed in previous theoretical and empirical research on the impact of minimum wage increases on the employment of a heterogeneous workforce in other sectors.

This study addresses the question of whether recent minimum wage increases impacted agricultural employment and whether the effect is different for US-born workers, immigrants, and guest workers. There are multiple institutional factors[a] that may lead to the high representation of immigrants in agricultural work. The domestic farm labor force is approximately 75% immigrants and half of the labor force may be undocumented (Zahniser et al., 2018). Farm employers have increasingly turned to the H-2A visa program, which allows guest workers to enter the U.S. for agricultural jobs on a temporary basis each year (Bampasidou & Michael, 2019). This heterogeneity in the farm workforce complicates analyses of minimum wage impacts on agriculture.

Several theoretical models demonstrate how minimum wage can have different impacts on native-born and immigrant workers, or a heterogeneous workforce more generally. Rising minimum wages could cause employers to shift employment from lower-skilled to higher-skilled workers, so if immigrants have (or are perceived to have) a different productivity level, US-born workers and immigrants could be impacted differently by the minimum wage (Orrenius & Zavodny, 2008; Zavodny, 2014).[b] Cadena (2014) argues that location decisions for low-wage workers are determined by expected wages, which rise with higher minimum wage but decline with lower probability of employment. If immigrants are more vulnerable to economic shocks,[c] they would face a higher penalty for a longer period of unemployment. Thus immigrants may be more likely to move to a low-wage area with greater chances of employment, return home, or move into another sector when the minimum wage rises. Further, immigrants may have lower bargaining power[d] with employers or, similar to Cadena (2014), face a higher penalty for a period of unemployment. Following this logic, undocumented immigrants may be less likely to see employment losses if the minimum wage increases. In contrast, workers with less bargaining power could be laid off before other workers. These differences between workers make it difficult to predict how minimum wage will influence agricultural workers; ultimately, this is an empirical question.

We use an empirical strategy that is based on spatial variation in the rapid introduction of minimum wage increases in several states around 2014–2015. A primary concern for empirical studies of minimum wage impacts is that economic and employment trends are related to both minimum wage increases and employment decisions. Recent minimum wage law changes are primarily driven by activism in the fast food industry and other service sector jobs, and democratic control in state governments (Ashby, 2017; DePillis, 2019), not agricultural labor or commodity markets. Under our empirical specification, these regulatory changes thus arguably provide exogenous variation to agricultural wages. Further, many empirical studies have shown that the farm economy has a negligible impact at most on nonfarm sectors, that is, Foster and Rosenzweig (2004), Hornbeck and Keskin (2014), and Weber et al. (2015). Our analysis covers 2006–2018, which incorporates years before and after major increase in minimum wage around 2014–2015. This period includes the most recent federal minimum wage change and covers the most recent publicly available H-2A (guest worker visa) data. Despite minimum wage increases being plausibly

[a]These could include: immigration laws that make it difficult for undocumented workers to advance, uneven enforcement of immigration law, recruiting and network effects (or lack thereof in other sectors), especially for new arrivals, the legacy of past migration programs, and fewer work opportunities in the home country (Martin & Calvin, 2010; Orrenius & Zavodny, 2009; Pfeffer & Parra, 2009; Weisbrot et al., 2017). Vijaya et al. (2015) also discuss how cultural norms can impact hiring practices and lead to intergroup inequality with regard to job sectors.

[b]Wu and Guan (2016) similarly assume different productivity levels for domestic agricultural workers and guest workers.

[c]For example, lower assets, higher liquidity constraints, or temporarily working in the United States.

[d]For example, if they are undocumented, filing a complaint could put them at risk.

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exogenous to farm labor markets, we cannot disprove the correlation of minimum wage increases with local economic and labor market conditions as well as employment trends across states. Cengiz et al. (2019) did not find strong evidence of pretrends or relevant concurrent trends during our study period but did not focus on the farm sector. We thus address this potential simultaneity by using a fixed effects specification[e] based on Addison et al. (2012, 2015) and Allegretto et al. (2011) that accounts for a broad range of local and regional economic conditions. Further, we conduct synthetic difference-in-differences analyses using only states that never increased their minimum wage and those that enacted a new law starting in 2014 or 2015 (estimated separately) following Arkhangelsky et al. (2021), to determine the potential for nonparallel pretrends to influence our analysis.

While a few datasets on farm employment are available, the American Community Survey (ACS) is the one that best allows us to address our research question. ACS microdata is collected annually, representative of the US population, contains information on immigration status, and is publicly available. The ACS includes unique information such as immigration status and recent migration history for both internal and international moves that allow us to analyze novel, policy relevant research questions, but it is not targeted to farm workers. While the ACS does not have information on hourly wages or exact number of weeks worked, it does contain information on the ranges of weeks worked, usual hours worked and annual earnings. The Census of Agriculture, which is conducted every 5 years, only provides data on the number of workers at the county level every 5 years and does not include currently unemployed workers. However, the aggregate number of hired farm workers reported in the Census of Agriculture (about 2.4 million in 2017) is remarkably comparable to the combination of total number of farm workers represented in the ACS and the total number of farm guest workers (or H-2A workers, which are excluded from the ACS), which is about 2.2 million.[f] We believe this similarity further substantiates the use of the ACS to study our research question, even though its use in agricultural economics research has been limited.

Similar to the Census of Agriculture, the National Agricultural Workers Survey contains rich information but is taken at the workplace, thus also excluding currently unemployed farm workers. The Quarterly Census of Employment and Wages (QCEW) is another option but does not have information on citizenship and immigration status. While the seasonality of some types of farmwork is a potential issue for analyzing minimum wage impacts on agricultural employment, the ACS asks workers if they were *looking for work in the last week*. Thus all analyses using ACS in this study only include employed individuals and those unemployed *and* looking for work. While this would not include seasonal workers that were sampled when they were in an "off-season"; the ACS is conducted throughout the year, so also captures seasonal farm workers. Any potential bias in our econometric analysis would be require both (1) systematically higher levels of "off-season" sampling in states with minimum wage increases and (2) large differences in levels of seasonal workers across states by minimum wage status. However, we believe that this type of sampling bias is unlikely and find that most states have substantial numbers of farm workers that have worked above and below a 40-week per year threshold (see Supporting Information: Figure C1). We further discuss this analysis, as well as the inherent tradeoffs between using different datasets that include farm workers in our Section 2.

Our study uses the ACS to estimate both individual and aggregate agricultural employment levels. We first estimate the impact of a minimum wage increase on the likelihood of an individual agricultural worker being employed, based on the self-identified job classifications provided by

[e] A difference-in-differences approach is not appropriate due to wide spatial dispersion in minimum wage increases during our study period.

[f] This figure reflects the number of farm workers between the ages 16 and 65 reflected in the ACS, calculated by totaling survey responses from all farm workers using the survey weights provided, and the number of guest workers surveyed in our US DOL data. While the farm workers in our ACS sample generally match the definition of hired labor used in the Census of Agriculture, the slightly different aggregate figures may be due to the Census of Agriculture including family workers of all ages, while we always exclude those under 16 and over 65 from our ACS estimates. Also, while the ACS works hard to include undocumented immigrants, this population may be particularly difficult to reach and therefore be underrepresented. The Census of Agriculture asks employers to report the number hired so nonresponse due to an individual's immigration status is less likely. Later in this manuscript, we will discuss how we exclude some ACS farm workers with high salaries and supervisory positions from our analysis, as such workers are typically not influenced by minimum wage changes.

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randomly selected ACS participants. A farm worker is defined as an individual that is either (1) currently working in agriculture or (2) if not employed but has been looking for work within the last week and agriculture was their primary occupation in the past 5 years. We further assess heterogeneous effects across groups by citizenship and immigration status. One challenge for our empirical strategy is that results may be influenced by workers leaving the farm sector. For example, if a farm worker is laid off, they will be observed as unemployed. However, the same worker may leave farming, find work in construction, and drop out of the sample. Analysis at an aggregated level allows us to determine if the total number of agricultural workers is declining in response to minimum wage increases. We use "PUMAs" or "Public Use Microdata Areas," as the ACS is designed to be aggregated to the PUMA level for local or regional analysis. PUMAs do not cross state or county boundaries and are designed to be used as the smallest geographical unit appropriate for statistical analysis in the ACS (see Supporting Information: Figures C2 and C3). Our analysis uses "consistent PUMAs (C. PUMAs)," as defined by the ACS as PUMAs with the same boundaries throughout our study period, since the PUMA boundaries have small changes once during our study period. However, we conduct all analyses with the earlier and later PUMA boundaries in robustness checks. We use ACS-supplied survey weights to aggregate individual data to the PUMA level, then estimate the impact of a minimum wage increase on the level of agricultural employment for native- and foreign-born domestic farm workers. This analysis captures all seasonal workers sampled within a year and also allows us to compare our results with the analysis of guest workers.

We estimate the impact of minimum wage on the number of H-2A workers at the county level, using data from the US Department of Labor (DOL) Office of Foreign Labor Certification on H-2A visa certifications at the employer level. Guest workers on this temporary visa round out the US farm labor force and these foreign workers may also be affected by changes in the minimum wage due to broader labor market dynamics. Guest workers are paid according to a different standard, the Adverse Effect Wage Rate (AEWR), which is effectively higher than minimum wage. We conduct several robustness tests related to assumptions around the construction of our sample and key variables, including using wages and hours as outcome variables. We also make use of a unique question on both international and internal migration in the ACS to study the effects of minimum wage on migration patterns of farm workers. This allows us to test if the employment trends we observe may be related to workers migrating across states or changing international migration patterns.

This work joins previous research (meta-analyses by Belman & Wolfson, 2014; Card & Krueger, 1995; Neumark & Wascher, 2007) that looks at the effects of minimum wage on employment, with a focus on heterogeneous effects in particular subpopulations, like teens, or within certain sectors, like retail. Findings are mixed but typically indicate no impact of the minimum wage on employment or a small negative wage elasticity (e.g., around −0.1 to 0.2 in some studies, see Neumark & Wascher, 2007). Cortes (2004) finds, using Current Population Survey data, that while immigrants are more likely to earn near the minimum wage, immigrants' wages rise at the same pace as domestic workers in industries with both high and low concentrations of immigrants. Recent research in the US farm sector used the Census of Agriculture and a long-differences specification to show that an increase in the minimum wage is associated with a decline in seasonal agricultural employment after 10–20 years, using data from 1992 to 2012 (Kandilov & Kandilov, 2020). Our study also complements other research that increases understanding of the dynamics of U.S. farm labor markets, including research that considers how citizenship status affects US farm work time allocations (Luo & Escalante, 2017) and earnings and welfare outcomes among farm and low-skill workers (Barham et al., 2020).

This study makes several contributions. First, we conduct a novel analysis of the differential short-term impact of the minimum wage on native-born and immigrant domestic workers, as well as H-2A workers. Second, we use data that covers the historically unprecedented increases in state minimum wages following the "Fight for 15" movement. Previous increases were almost always incremental or relatively small and thus their impact may be more difficult to empirically disentangle from broader farm sector trends. Third, the ACS data allows us to investigate the impact

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of the minimum wage on employment, as well as providing additional information on annual earnings, hours worked, and migration. This data set may be useful for future analyses of agricultural labor markets. Fourth, our study adds to the recent research on minimum wage and agriculture, including studies on minimum wage, piece rate, and productivity (Hill & Burkhardt, 2021; Richards, 2020), which suggest that increases to minimum wage and piece rates may not necessarily alleviate agricultural labor shortages.

We find that increasing the minimum wage may have a small, negative short-term impact on the likelihood of being employed as a farm worker; a $1 increase leads to around a one percentage point decrease in the likelihood of employment. This net effect masks substantial heterogeneity in how minimum wage changes impact workers by citizenship and immigration status. Specifically, for noncitizen farm workers, an increase in minimum wage decreases their likelihood of employment less than that of citizens, by about 0.9 percentage points. However, we find no evidence for a relationship between the aggregate level of domestic employment of agricultural workers and guest workers and minimum wage levels. Overall, consistent with prior research on the effect of minimum wage on employment in other sectors, we do not find strong evidence that minimum wage increases decrease agricultural employment for domestic or H2-A workers, or that employers would lay off domestic workers in favor of guest workers. The ACS data does allow us to examine additional questions regarding earnings, migration, and employment impacts on the intensive margin. We do find that increasing the minimum wage increases annual earnings for noncitizen farm workers, suggesting the wage floor may be binding specifically for these workers. We also find that noncitizens are less likely to move to states with higher minimum wages, but this effect is likely driven by new arrivals to the United States rather than internal migration within the United States. Finally, there is also no evidence of an effect on hours worked, supporting our other findings on the impact of the minimum wage on agricultural employment.

The subsequent section describes the data, while section three covers the empirical strategy of the paper. Section four presents and discusses the results, including several robustness checks. The final section concludes with a summary of the findings and questions for future research. For the interested reader, in Supporting Information: Appendix A we summarize farm labor trends and institutional details, including minimum wage law in the United States and the H-2A visa.

2 | DATA

The primary data used in this study comes from the ACS and US DOL Office of Foreign Labor Certification's disclosure data regarding H2-A guest workers. The first key data set comes from the IPUMS US-Microdata extracts of the ACS from 2006 to 2018 (Ruggles et al., 2020). This data contains an individual-level, representative sample of US residents, with detailed information on their employment status and job details, including employment in agriculture. This data also includes information on nationality and immigration status, so US-born citizens and immigrants can be identified. The sampling method covers both authorized and unauthorized immigrants, but it does not identify immigration status beyond "foreign born" and "naturalized citizen," so we cannot consider authorized and unauthorized immigrant workers separately. Approximately 13.2% of the sample is foreign born and immigrants make up 14.4% of the US population, so in this respect the ACS reflects the US immigrant population (Budiman, 2020).

US Census Bureau survey data (such as the ACS) is designed to capture the resident population and thus does not specifically attempt to survey the nonimmigrant population, which includes temporary visa holders (Greico, 2006). Nonetheless, nonimmigrants may be surveyed and included in the noncitizen group of foreign-born individuals. We do not believe the ACS includes a meaningful population of H-2A workers, as this group is difficult to enumerate in any survey (Hertz, 2019; Migration Policy Institute, 2020). Additionally, the ACS is conducted at various points throughout the year and the short duration of most workers' stays makes it even more unlikely an

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H-2A worker would actually be surveyed, as opposed to longer-term nonimmigrants such as foreign students. In their attempts to estimate the unauthorized immigrant population by removing lawful noncitizens from the ACS data Passel and Cohn (2016) account for many temporary work visas but not H-2A visas, suggesting these might not be of substantial concern. Given the legal requirement to provide housing for H-2A workers (which is often dorm-style), any H-2A worker in the survey is more likely to appear in a group quarter setting. In several robustness checks, we exclude observations of individuals living in group quarters, with largely consistent results.

We limit our primary analysis of ACS data to those in the labor force who indicate they work in a nonsupervisory, agricultural job. The ACS determines a worker's industry (and occupation, which is how we determine a nonsupervisory role) based on the main job they worked the week prior or, if they are unemployed but looking for work, the main type of work they did in the 5 years prior. We further exclude logging and mining workers, as well as produce delivery and bookkeeping workers. Last, we drop anyone who reports a personal income over 100,000 dollars a year, as these workers are likely quite different from the domestic and H-2A workers in the rest of the sample and often appear to have another job and work in agriculture on the side. Further, Cengiz et al. (2019) find that higher wage workers are unlikely to be affected by minimum wage increases. This gives us about 113,000 usable observations that, based on survey weights, are representative of approximately 11.5 million farm workers in the United States over the period of study. These workers are most likely to be affected by minimum wage increases. These restrictions do decrease our number of observations from about 290,000 in the full farm worker sample, so we include robustness checks using the full sample, where the main difference is driven by including supervisors.[g] Our main analysis sample most likely reflects workers similar to H-2A workers. However, we also estimate our primary model with a sample that limits the data to only include nonsupervisory workers in the crop sector. More details on the construction of demographic control variables are provided in Supporting Information: Appendix A.

ACS data provides information on additional outcomes beyond employment: annual wage income, usual weekly hours worked, and whether the individual moved to their current state in the last year. For movers, we create an indicator variable equal to one for someone who moved from another US state or country to their current state in the last year. We recognize the limitations of using the ACS data, as it is not designed specifically to capture the agricultural workforce and surveys are conducted throughout the year, while other agricultural worker surveys are more careful to survey workers during periods of high agricultural employment such as harvest season. Despite the limitations, the ACS data is valuable to this research as it contains information on citizenship and immigration status, recent migration, wages, and hours worked, that is not all available in other surveys. ACS also includes unemployed farm workers. Further, as discussed in the introduction, estimates of the number of farm workers using ACS data are comparable to the Census of Agriculture. Other surveys, such as the National Agricultural Workers Survey, conduct interviews at job sites, meaning all those surveyed are employed and we would not be able to assess changes in the likelihood of employment with that data. We assume that people who indicate that they are agricultural workers but are currently unemployed and looking for work are seeking employment in the agricultural sector or have most recently lost a job in that sector.

The ACS asks if respondent is currently looking for work, and hence our analysis should exclude seasonal workers that are off-season and not looking for work. Because the ACS is conducted throughout the year, it also captures currently employed seasonal workers. However, we believe it is unlikely that a farm worker in the ACS reports that they are looking for work because they are interviewed out of season, when they are actually *not* looking for work at that time. We further do not expect systematic bias that could occur, for example, if surveys were sent to high minimum wage

[g]To compare to the 2017 estimates provided in the introduction, using survey weights, we drop from about 2.2 million farm workers to 1.1 million nonsupervisory workers represented in our main analysis sample.

states always during the off-season or otherwise sampled differently by minimum wage status. We also find no evidence that individual states are dominated by fulltime or seasonal farmworkers. Supporting Information: Figure C1 shows the distribution of farm workers reporting more and less than 40 weeks worked in the prior year. There are substantial numbers of workers below and above this threshold in each state, with the possible exception of California, where seasonal labor seems slightly overrepresented. California did increase its minimum wage in our survey period, but the prevalence of seasonal workers in California does not increase when the minimum wage goes up. Altogether, this suggests that main results should not be biased by seasonality.

For PUMA-level analysis, we use the ACS geographical information to tabulate the number of nonsupervisory agricultural workers surveyed in a consistent PUMA and year. C. PUMAs are statistical geographic areas with at least 100,000 people, nested within states, geographically contiguous and cover the entire United States (U.S. Census Bureau, 2011). A PUMA can be smaller or larger than a county. PUMAs without farm workers are not included in our sample. We also calculate PUMA-level characteristics using the ACS microdata, including fraction of noncitizens and naturalized citizens working in agriculture, and the overall PUMA unemployment rate.

PUMA delineation is adjusted every decade to match population changes, but there is substantial overlap between previous and current PUMAs. Our main analysis uses C. PUMAs, which are aggregations of 2000 and 2010 PUMAs that create geographical boundaries that match at least 99% across the two periods (IPUMS, n.d.). The general overlap of these geographic units and visual analysis of our data shows that almost all the observations from our sample that are in the "same" PUMA[h] in 2011 and 2012 (the year the PUMA delineations were adjusted) are in the same C. PUMAs, suggesting at least geographic proximity. Total nonsupervisory agricultural workers per year from the ACS data are reported in Supporting Information: Table A1. We estimated specifications assuming PUMAs were the same for the entire study period and also split the data and analysis into two periods, so that PUMAs were consistent within each subsample, with largely consistent results.[i] Maps of PUMA delineations are available in Supporting Information: Figures C2 and C3. For our primary analysis using ACS data, the smallest level of aggregation will be the C. PUMA. To keep analysis consistent with guest worker analysis, we only include C. PUMAs that have farm workers in the main analysis.

The DOL data contains information on all H-2A hires from 2006 to 2018 at the employer level. It includes detailed information such as employer name and address, the number of workers requested and the number certified, job type, job duration, and more. We use this data from 2006 to 2018. The last change in the federal minimum wage occurred in 2009, so many states that do not set their own minimum wage rate have had a constant rate since 2010. US territories and Washington DC are dropped, as well as addresses from Alaska and Canada.[j] Each state's AEWR is added to this data set. The data is aggregated to the county level using the employer address, which is provided for every observation. County-year level summary statistics for H-2A workers are reported in Supporting Information: Table A2. We conduct robustness checks with specifications that aggregate the data using the work site address, with largely consistent results.

Minimum wage data is drawn from the Federal Reserve Bank of St. Louis Economic Data U.S. DOL (nd) and covers all 50 states from 2006 to 2018. Many states set their state minimum wages equal to the federal minimum wage, which is $7.25 for most of the period. Five others have no official state minimum wage and thus only the federal minimum wage covers workers in these states, so we assign these states the federal minimum wage as well. The remaining states have all set minimum wage above the federal minimum wage sometime during the sample period. A full description of state-level minimum wages throughout the time period is provided in Supporting

[h] For example, Alabama PUMA 100 in 2011 might not be drawn the same as Alabama PUMA 100 in 2012.

[i] Although excluded for space considerations, these results are available upon request.

[j] These amount to less than 0.5% of the total data set.

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Information: Appendix A. County-level data from the Bureau of Economic Analysis (BEA) on farm employment and agricultural cash receipts U.S. Bureau of Economic Analysis (n.d.) (in millions) as well as county and state unemployment rate data from the Bureau of Labor Statistics (BLS) supplements this data set U.S. Bureau of Labor Statistics (n.d.).

Our empirical strategy leverages variation in minimum wage levels, H-2A hires, and domestic farm worker employment levels across time and space. Employment and control variables used in our analysis of domestic farm workers are reported in Table 1, while data used in the analysis of guest worker employment in reported in Table 2. The minimum wage was increasing over our study period, with substantial variation between states (Supporting Information: Figure C4). Supporting Information: Figure C5 shows the increase in H-2A hiring over time, while Supporting Information:

TABLE 1 Domestic summary statistics PUMA level

Variable	Mean	Min.	Max.	PUMAs represented
Number of workers in C. PUMA	1184	2	40364	9707
Individual level				
Variable	**Mean**	**Min.**	**Max.**	**Pop. represented**
Percent of farm workers employed	87.48	0	100	1.150e + 07
State unemp. rate	6.659	2.400	13.70	1.150e + 07
Percent not US citizen	47.19	0	100	1.150e + 07
Percent naturalized	5.585	0	100	1.150e + 07
Percent foreign born	52.77	0	100	1.150e + 07
Percent female	23.94	0	100	1.150e + 07
Percent in crop production	63.56	0	100	1.150e + 07
Usual hours worked/week	40.76	0	99	1.150e + 07
Percent moved in last year	4.374	0	100	1.150e + 07
Percent moved within US last year	2.142	0	100	1.150e + 07

Note: Summary statistics for sample of all nonsupervisory agricultural workers surveyed in the ACS between 2006 and 2018. Data were weighted using weights provided by IPUMS in the survey. Average values in full sample of all agricultural workers. The fraction of workers employed measures the fraction of people who report their occupation is agricultural work who also reported being employed at the time of survey. The state unemployment rate is measured as a percentage.

Abbreviations: ACS, American Community Survey; C. PUMA, consistent PUMA; PUMAs, Public Use Microdata Areas.

TABLE 2 H-2A summary statistics

Variable	Mean	Std. Dev.	Min.	Max.	*N*
Number of H-2A workers certified in county	110.85	482.038	1	12,093	16,703
County-level unemployment rate	5.971	2.796	1.1	28.9	16,700
County-level cash receipts	160.966	333.705	0.435	5934.863	16,440
County-level farm compensation (thousands)	14,379.936	44,611.637	61	812,671	16,440
County-level farm employment	1083.818	1534.36	17	21,243	16,440
County per capita income	39,216.688	11,607.256	16,513	190,997	16,501

Note: Average values across counties and years 2006–2018 for variables used H-2A analysis. The county-level unemployment rate is measuredas a percentage.

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Figure C6 shows greater stability but with notable recent declines in domestic farm workers reporting in the ACS. These aggregates mask significant across-state variation in farm employment levels. Supporting Information: Figure C7 presents the variation in the total number of H-2A workers certified from 2006 to 2018 for each state (except Alaska, which is excluded from all analysis here). Likewise, Supporting Information: Figure C8 presents the comparable variation in domestic farm workers surveyed in the ACS over the same time period.

3 | EMPIRICAL MODEL

The literature on the theoretical and empirical impacts of raising the minimum wage predicts ambiguous effects of minimum wage on immigrants and US-born farm workers. For the interested reader, we provide an in-depth discussion of this literature in Supporting Information: Appendix A. The approach of this study is to first establish the average effect of minimum wage on all farm workers, followed by farm workers by immigration status. We then explore different mechanisms that might be at play and conduct a series of robustness checks. Our strategy measures the short-term impact of the minimum wage on employment in agriculture, accounting for the diversity of the farm labor force in terms of citizenship and immigration status. However, in some cases, minimum wage may change long-term employment dynamics (see Kandilov & Kandilov, 2020; Meer & West, 2016), which we would not capture. Our identification strategy hinges on the assumption that minimum wage changes are not jointly determined with agricultural labor markets, as discussed in the Introduction.

The first empirical model introduced in Equation (1) is a standard fixed effect approach using area (either state or C. PUMA) fixed effects and controlling for a linear time trend. This model follows Addison et al. (2012, 2015); the fixed effect approach allows us to use data from the entire United States. Addition of linear time trends, similar to (Addison et al., 2012) and as suggested in Allegretto et al. (2011), accounts for state-level differences in agricultural employment trends.

$$Employed_{it} = \alpha_0 + \beta\ Minwage_{st} + \alpha_t + \alpha_p + \gamma_1 X_i + \gamma_2\ state \times year + E_{ist}. \tag{1}$$

The outcome is a binary indicator for whether an individual (i), who works in an agricultural role in year t, reported being employed. Individuals are treated as unemployed if they not currently employed and were looking for work in the past week. The mean of this variable in a state and year would be the state-level employment rate[k] in agriculture. This linear probability model assesses the impact of the minimum wage on the probability that an individual in agriculture is employed. Errors for all specifications are clustered at the state level, since treatment is assigned at this level (Abadie et al., 2017; Cameron & Miller, 2015). We also weight regressions using the sampling weights provided in the ACS data.

β is the coefficient of interest and indicates the effect of a \$1 increase in the minimum wage on the likelihood an individual reports being employed in agriculture. We also use the natural log of minimum wage as an explanatory variable. X_i includes person-specific control variables: immigration status classification, race/ethnicity, age, marital status, and gender. Immigration status is the main point of interest in this study, but factors including individual and structural discrimination, institutional characteristics, and network connections, affect employment outcomes for workers based on their social identities. We, therefore, control for some of these characteristics available in the ACS data (see Supporting Information: Appendix A for further discussion of control variables and the use of demographic controls such as race and gender). Year (α_t) and PUMA (α_p) fixed effects are included, as well as a state-by-year linear trend ($\gamma_2\ state \times year$). We use C. PUMAs to define the local area for observation in all primary specifications. We also use state

[k]Employment rate is the percentage of the labor force (employed individuals plus individuals looking for work) that is currently employed.

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fixed effects instead of PUMA fixed effects in one specification, which increases identifying variation in our model but relaxes controls for local labor market conditions.

To test whether the effect is different for immigrants, we interact the minimum wage with an immigrant status indicator. The ACS provides detailed information on nationality as well as some information on citizenship and immigration status. While we do not observe whether an individual is an unauthorized worker, we can tell who reports being a naturalized citizen. Thus, we further separate immigrant workers by naturalization status when using a linear probability model to estimate the following equation.

$$Employed_{it} = \alpha_0 + \beta_1\, Minwage_{st} + \beta_2\, Minwage_{st} \times (non\text{–}citizen) + \beta_3\, Minwage_{st} \times (naturalized) + \alpha_t + \alpha_p + \gamma_1 X_i + \gamma_2\, state \times year + E_{ist}. \tag{2}$$

We conduct several robustness checks that address underlying assumptions about variable construction and model composition. We estimate our robustness checks with a state-level unemployment control. While unemployment could be jointly determined with minimum wage increases, it is also a relevant measure of local economic conditions that has been used as a control in other studies of industry-specific minimum wage impacts, such as Addison et al. (2012).[1] First, we estimate our model that uses a sample of all farm workers (including managers) and only crop workers, which may be more comparable to guest workers. In one specification, we also exclude farm workers living in group quarters. While this lowers the likelihood that respondents are guest workers, it also likely excludes some workers that earn minimum wage or near-minimum wage. Another robustness check uses a PUMA time trend, instead of a state time trend. Finally, we treat respondents who identify as farm workers and report not being in the labor force (NILF) as unemployed, followed by estimating Equation (2) using NILF as the outcome. This analysis relaxes our assumptions about employment status and sheds light on whether some individuals may have left the workforce in response to minimum wage increases. These individuals may also be seasonal farm workers that are "off-season," and this serves as a robustness check that we are capturing such workers in our analysis.

Since the ACS is not a panel, there may be concerns that laid-off workers drop completely out of the labor market and thus are not counted as unemployed. Further, some workers may move to employment in other sectors. Immigrant workers might choose to leave the United States or move to another state in search of a job, as they may be less attached to a particular location than unemployed native workers. To address this, and conduct similar analysis as is done with the H-2A sample, we aggregate the number of employed agricultural workers in the ACS each year by state and C. PUMA. This method will also capture workers who may leave the sector as wages rise in other (or all) sectors.

This aggregation gives a count of the number of agricultural workers surveyed in each local area, each year. Both job losers and leavers are reflected by lower levels of overall workers. With this analysis, we cannot separate out these two groups, but we can interact the minimum wage with the fraction of farm workers who are naturalized citizens and other immigrants in the area. We use both the count of workers and the natural log of the count of workers as dependent variables, to address the skewness of the data. We estimate Equation (3) using ordinary least squares (OLS), with the treatment variable being the state minimum wage that covers the region. We also estimate the model using the natural log of minimum wage. As in other specifications, we include a state-by-year linear trend as well as time-varying local characteristics such as the PUMA unemployment rate.

$$ln(workers_{pt}) = \alpha_0 + \alpha_p + \alpha_t + \beta\, minwage_{st} + \gamma_1 X_{pt} + \gamma_2\, state \times year + E_{pst}. \tag{3}$$

[1]Results from any of our primary specifications and robustness checks do not change with inclusion or exclusion of unemployment levels or trends at the county or state level. While we include some relevant robustness checks, the additional alternative results are available from the authors upon request due to space considerations.

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Again β is the coefficient of interest, with the same fixed effects and linear trends as Equation (1). Similar to Equation (2), we add specifications that include interactions of the minimum wage and share of noncitizens within each PUMA (min. wage × % noncitizen) and naturalized citizens (min. wage × % noncitizen). In this model, control variables X_{pt} reflect C. PUMA level characteristics such as percentage Hispanic, percent immigrant, and C. PUMA overall unemployment rate, since the data is at the C. PUMA level. Similar to the individual model, we conduct robustness checks on sample composition and time trends. Supporting Information: Appendix D includes a synthetic difference-in-differences analysis following Arkhangelsky et al. (2021), to address the possibility that our results are driven by nonparallel pretrends. The synthetic difference-in-differences analysis builds on insights from the synthetic control literature, but systematically weights control observations to generate a control with the best parallel pretrends, rather than identical preperiod trends. This supplemental analysis is conducted as a robustness check, as a difference-in-differences methodology is not perfectly suited to our setting. We also account for sampling weights provided in the ACS when aggregating data to the PUMA level so that each PUMA-year observation is representative of the number of farm workers in the area, rather than reflecting changes in ACS sampling practices.

Next, we investigate how the minimum wage impacts guest workers. We test the predictions discussed above using H-2A certifications from 2006 to 2018. In this case, we estimate the following equation using OLS.

$$ln(H2A_{ct}) = \alpha_0 + \alpha_c + \alpha_t + \beta\, minwage_{st} + \gamma_1 X_{ct} + \gamma_2 state \times year + E_{cst}. \tag{4}$$

The outcome in this model is the natural log of county-level H-2A hires and we include county and year fixed effects, as well as time-varying county characteristics such as unemployment rate to account for changes in the labor market and farm sector over time. Charlton and Castillo (2021) found state unemployment levels to be negatively related to H-2A demand. We also include state-specific time trends to account for potentially different economic trends at the state level. Control variables X_{ct} are additional county-level control variables, time-varying controls such as unemployment rate and farm cash receipts. Because unemployment and cash receipts are potentially jointly determined with employment levels, we interpret specifications with these controls as potentially affected by endogeneity.

We conduct several robustness checks, including adding additional control variables, county instead of state time trend, using worksite instead of employer address, and a synthetic difference-in-differences approach. We also explore the possibility that the relationship between AEWR and minimum wage could influence labor demand, as suggested by Wu and Guan (2016). To do this, we use the AEWR-Minimum wage gap as our treatment variable, albeit with the acknowledgment that AEWR could be jointly determined with guest worker demand. If the gap between the minimum wage and the AEWR is large, then domestic (non-H2-A) workers are more advantageous from the employers' perspective as they are relatively cheaper than H2-A workers, compared to areas where the gap between the minimum wage and AEWR is small and there is a smaller difference in wages between the two types of workers. A negative coefficient on the wage gap measure suggests that the larger the AEWR-minimum wage gap, the fewer H2-A work a farm employer would use presumably because the advantage to using domestic workers is higher in that state. We also use lagged minimum wage as an explanatory variable in one specification, as the current-year AEWR may have a stronger relationship with the previous year's minimum wage.

This analysis is different from than analysis with ACS data, as the H-2A data reflects just the labor demand side. H-2A hiring is driven by US demand for this labor and there appears to be a consistent supply of potential H-2A workers. This assumption is supported by continued immigration from Mexico and Central America as well as recent political dealings between the US government and foreign governments. In 2019, President Trump agreed to triple the number of H-2A visas awarded to Guatemalans as part of an immigration policy agreement to hold asylum-seekers in Guatemala (Menchu, 2019). This suggests that there is ample demand for these visas so the number of workers hired reflects the US-based demand for such workers.

Finally, we conduct two additional analyses that provide internal validity for how minimum wage influences agricultural employment, as well as insight into the underlying drivers of net employment impacts. We first estimate Equation (2) using annual wage income and hours per week as the outcome variables. For annual wage income, we control for usual hours worked, which allows us to better isolate whether an increase in the minimum wage actually increased wage levels. As in other specifications, we vary the level of fixed effects and time trends. We also estimate Equation (2) using migration (including whether this migration was from inside the United States) as the outcome, with standard controls and fixed effects. Further, we are able to consider minimum wage in the previous state worked in (if it existed) as a treatment variable, to see if this induced migration between states.

4 | RESULTS

The estimation results of Equations (1) and (2) are presented in Table 3 and provide evidence that an increase in the minimum wage has a small, negative impact on the likelihood of employment for agricultural workers, though this impact may be smaller for noncitizen workers. The main specification is in column (1), while column (4) includes an interaction term to capture the potentially heterogeneous effects of the minimum wage by immigration status. The net effect on all workers presented in columns (1)–(3) are statistically significant at the 5% test level and suggests a decrease in likelihood of employment of less than 1% percentage point for each dollar increase in the minimum wage. When we use the log of the minimum wage (column 3), we find that a 1% increase in the minimum wage is associated with approximately a 0.06 percentage point decline in the likelihood of being employed; changes in the minimum wage appear to have at most a small impact on overall employment in agriculture. We cannot distinguish effects between native-born US citizens and naturalized citizens, but the noncitizen group experiences a relatively smaller net effect (column 4). Under this specification, domestic workers experience a statistically significant decrease in employment when the minimum wage increases; slightly over 1 percentage point for each dollar increase, but the impact is concentrated among native-born and naturalized US citizens. This difference could reflect several potential characteristics of noncitizens, including higher productivity and lower mobility between sectors.

These results do not hold in a sample of all agricultural workers (Supporting Information: Table B1), which is expected given that managers are included in this much-larger group. Results are largely consistent for a smaller group of nonsupervisory crop workers (Supporting Information: Table B2), excluding group quarters (Supporting Information: Table B3), excluding a control for unemployment (Supporting Information: Table B4),[m] controlling for PUMA linear trends (Supporting Information: Table B5), and treating workers who are not in the labor force as unemployed (Supporting Information: Table B6). We also find no evidence that minimum wage influences the decision to exit the labor force (Supporting Information: Table B6, columns 3 and 4). Most of these specifications also show evidence of heterogeneous treatment effects. Potential explanations for this result will be explored in subsequent analyses. The primary finding is that there is a small impact of minimum wage on the likelihood of employment for domestic farm workers, with evidence of different effects for noncitizen farm workers.

Results using data that aggregate employment levels at the PUMA (or C. PUMA) level are reported in Table 4. We find that increasing the minimum wage has a statistically insignificant relationship with the net level of employment in the sample (columns 1–6) and by citizenship and

[m]Due to concerns of the potential endogeneity of the unemployment controls, we present specifications without this control for the guest workers analysis (Table 5 column 5) and domestic workers individual model (Supporting Information: Table B4) and results are consistent with main findings. We similarly replicated the analysis for all other main tables (Tables 4, 6, and 7) without the unemployment control and again found consistent results. We do not include those results for the sake of brevity but they are available upon request.

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TABLE 3 Individual model, nonmanager sample

	(1) Employed = 1	(2) Employed = 1	(3) Employed = 1	(4) Employed = 1
Minimum wage	−0.00867*	−0.00826*		−0.0131
	(0.00361)	(0.00387)		(0.00359)***
Not a US citizen	0.0769***	0.0797***	0.0769***	0.00476
	(0.00727)	(0.00898	(0.00727)	(0.0299)
Naturalized US citizen	0.0493***	0.0495***	0.0493***	−0.0219
	(0.00823)	(0.00936)	(0.00823)	(0.0484)
State unemployment rate	−0.00561**	−0.00570**	−0.00559**	−0.00583**
	(0.00191)	(0.00181)	(0.00198)	(0.00196)
Ln(min. wage)			−0.0612	
			(0.0275)*	
Min. wage × noncitizen				0.00893**
				(0.00332)
Min. wage × naturalized				0.00883
				(0.00543)
Observations year FE	112,944	112,973	112,944	112,944
Year FE	Yes	Yes	Yes	Yes
State FE	No	Yes	No	No
C. PUMA FE state time trend	Yes	No	Yes	Yes
C. PUMA time trend	No	No	No	No
Demographic controls	Yes	Yes	Yes	Yes

Note: Standard errors in parentheses. The sample includes nonmanager agricultural workers surveyed in the ACS between 2006 and 2018. Demographic controls include dummies for sex, Hispanic ethnicity, 9 indicators for race, indicators for marriage status, and age (see Supporting Information: Appendix A for a discussion of demographic variables). Data from all states except Alaska are used in each specification. Columns 1 and 2 present the main specification from Equation (1); column 1 controls for (consistent) PUMA level geographic fixed effects, rendering any state-level fixed effects redundant, while column 2 does not control for PUMA level fixed characteristics and only includes state-level geographic fixed effects. Columns 3 presents specifications similar to column 1 but the natural log of minimum wage. Column 4 presents the estimation of Equation (2), the main specification with interactions with immigration status. The number of observations may fluctuate slightly due to the use of different control variables but this fluctuation is typically less than 0.5% of the sample. All regressions are weighted using survey sampling weights and standard errors are clustered at the state level.

Abbreviation: ACS, American Community Survey.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

immigration status. Columns 1 and 2 estimate the impact of a dollar increase in the minimum wage, while columns 3 and 4 use the natural log of the minimum wage as the independent variable. As such, column 3 presents an approximate elasticity and column 4 shows that a 1% increase in the minimum wage is associated with a decrease of 423 workers (though this is statistically insignificant). Columns 5 and 6 show results with an interaction between the treatment variables and the fraction of agricultural workers who are immigrants.

PUMA-level results are largely consistent with excluding group quarters (Supporting Information: Table B7), including only nonsupervisory crop workers (Supporting Information: Table B8), and inclusion of PUMA time trends (Supporting Information: Table B9). While in some

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TABLE 4 PUMA level model

	(1) Ln(number of local workers)	(2) Local workers	(3) Ln(number of local workers)	(4) Local workers	(5) Ln(number of local workers)	(6) Local workers
Minimum wage	−0.0289	−65.11			−0.00329	−63.82
	(0.0256)	(51.66)			(0.0281)	(55.32)
Not a US citizen	0.114*	142.5***	0.113*	142.2***	0.801*	64.11
	(0.0552)	(28.38)	(0.0552)	(28.45)	(0.350)	(243.0)
Naturalized US citizen	−0.0195	−14.12	−0.0196	−14.25	0.572	389.0
	(0.0511)	(28.20)	(0.0512)	(28.38)	(0.328)	(227.4)
Unemployment rate	0.454	666.8	0.458	674.3	0.420	668.2
C. PUMA	(0.502)	(483.6)	(0.500)	(488.9)	(0.510)	(485.4)
Ln(min. wage)			−0.186	−423.2		
			(0.196)	(369.0)		
Min. wage × % noncitizen					−0.0892	10.17
					(0.0478)	(29.46)
Min. wage × % naturalized					−0.0765	−51.56
					(0.0400)	(28.33)
Observations	9671	9671	9671	9671	9671	9671
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	No	No	No	No	No	No
C. PUMA FE	Yes	Yes	Yes	Yes	Yes	Yes
State time trend	Yes	Yes	Yes	Yes	Yes	Yes
C. PUMA time trend	No	No	No	No	No	No
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: The sample includes all those who indicate their job description would be nonmanagerial farm work, regardless of actually being employed at the time of the survey (excludes NILF). These workers are then aggregated to the PUMA-level to form the unit of observation, accounting for sampling weights. For all analysis, C. PUMA across 2000 and 2010 delineation. All specifications control for the proportion of PUMA that is Hispanic though coefficients are consistently small and insignificant on this control (see Supporting Information: Appendix A for a discussion of demographic variables). Data from all states except Alaska is included. Columns 1–4 present main specifications of Equation (3), while columns 5 and 6 interact minimum wage with the proportion of noncitizens and naturalized citizens in the PUMA. Standard errors are clustered at the state level.

Abbreviations: C. PUMA, consistent PUMA; NILF, not being in the labor force; PUMAs, Public Use Microdata Areas.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

robustness checks, we find evidence of heterogenous effects, these are inconsistent across specifications. Similar to the individual analysis, we do not find evidence of heterogeneous impacts when including supervisory farm workers (Supporting Information: Table B10) in our sample. Further, results are largely consistent with using different assumptions about PUMA construction and time period but are most similar to our primary results using 2012–2018.[n] Similarly, we find no negative impact on employment level using the synthetic difference-in-differences specification,

[n]These results have been excluded due to space considerations but are available upon request.

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TABLE 5 H-2A model

	(1) Ln(number of H-2A workers)	(2) Ln(number of H-2A workers)	(3) Ln(number of H-2A workers)	(4) Ln(number of H-2A workers)	(5) Ln(number of H-2A workers)
Minimum wage	0.0559	0.0647*			0.0619
	(0.0312)	(0.0295)			(0.0328)
County unemployment rate	−0.0239**	−0.0247***	−0.0237***	−0.0244***	
	(0.00682)	(0.00682)	(0.00679)	(0.00678)	
County-level cash receipts		0.000142		0.000138	
		(0.000147)		(0.000146)	
Ln(min. wage)			0.407	0.469*	
			(0.226)	(0.213)	
Observations	16,419	16,161	16,419	16,161	16,422
Year FE	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
State time trend	Yes	Yes	Yes	Yes	Yes

Note: Standard errors in parentheses. Data from counties in all states except Alaska are used. Columns 2 and 4 control for county-level farm cash receipts to account for differences in farm success across counties. Columns 1 and 2 use the minimum wage as the wage variable of interest, while columns 3 and 4 use the log of the minimum wage to measure the elasticity. Column 5 presents the results without county-level fixed effects but not other controls. The number of observations may fluctuate slightly due to the use of different control variables but this fluctuation is typically less than 0.5% of the sample. Standard errors are clustered at the state level.

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

which is constructed at the state level rather than the PUMA level and has the benefit of weighting the control group to establish stronger parallel pretends (Supporting Information: Table D2). For details on this analysis including the specific treated and control states, the standard error calculation, and the estimation package, see Supporting Information: Appendix D. This finding is consistent with most major minimum wage changes occurring from 2014 onward.

We next estimate the impact of the minimum wage increase on the demand for guest workers. Results from estimation of Equation (4) are reported in Table 5. We do not find a robust, consistent relationship between the number of guest workers and minimum wage increases. While the minimum wage coefficient is statistically significant when farm cash receipts (columns 2 and 4) are included as a control, this may be due to the correlation of receipts with local farm labor demand. Overall, we do not find evidence that minimum wage increases are influencing guest worker demand. This suggests that there are no domestic farm workers laid-off to hire guest workers and is consistent with previous results of no or very small changes in employment levels, as well as broader labor shortages in agriculture.

These results do not meaningfully change when using additional county-level characteristics (Supporting Information: Table B11), adding a county time trend (Supporting Information: Table B12), using raw number of certifications (Supporting Information: Table B13), or H-2A worksite instead of employer address (Supporting Information: Table B14). Further, our results do not change when we use alternative measures of wages (Supporting Information: Table B15).[o] The

[o]Due to concerns of the potential endogeneity of the unemployment controls, we present specifications without this control for the guest workers analysis (Table 5 column 5) and domestic workers individual model (Supporting Information: Table B4), and results are consistent with main findings. We similarly replicated the analysis for all other main tables (Tables 4, 6, and 7) without the unemployment control and again found consistent results. We do not include those results for the sake of brevity but they are available upon request.

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results are also consistent with the synthetic difference-in-differences specification (Supporting Information: Table D2). Lagged minimum wage, which may be more likely to influence AEWR than the current minimum wage, does not have a statistically significant relationship with guest worker demand. While the gap between minimum wage and AEWR hypothetically could influence hiring,

TABLE 6 Wage and hours results

	(1) Annual wage income	(2) Annual wage income	(3) Hours worked/week	(4) Hours worked/week	(5) Annual wage income	(6) Hours worked/week
Minimum wage	25.52	−4.042	−0.204	−0.132	11.30	−0.128
	(172.5)	(167.2)	(0.199)	(0.210)	(164.8)	(0.218)
Min. wage × noncitizen	266.8*	296.7**	−0.0506	−0.122	230.1*	−0.138
	(108.1)	(95.52)	(0.147)	(0.119)	(101.2)	(0.128)
Min. wage × naturalized	−261.70	−194.3	−0.172	−0.156	−218.6	−0.173
	(107.8)	(101.40)	(0.212)	(0.229)	(103.20)	(0.251)
Usual hours worked per week	323.2***	320.3***			319.5***	
	(6.938)	(7.197)			(7.304)	
Not a US citizen	−2388.9*	−2726.3**	2.934*	3.337**	−2187.7*	3.466**
	(943.7)	(828.6)	(1.247)	(1.031)	(881.1)	(1.119)
Naturalized US citizen	2908.8**	2264.7**	2.843	2.791	2486.0**	2.919
	(924.9)	(843.8)	(1.437)	(1.524)	(876.3)	(1.659)
State unemployment rate	−361.9***	−348.6***	−0.0913	−0.0656	−366.7***	−0.0866
	(92.45)	(90.47)	(0.126)	(0.130)	(88.00)	(0.135)
Observations	112,973	112,944	112,973	112,944	112,944	112,944
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	No	Yes	No	No	No
C. PUMA FE	No	Yes	No	Yes	Yes	Yes
State time trend	Yes	Yes	Yes	Yes	No	No
C. PUMA time trend	No	No	No	No	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors in parentheses. The sample includes nonmanager agricultural workers surveyed in the ACS between 2006 and 2018. Demographic controls include dummies for sex, Hispanic ethnicity, 9 indicators for race, indicators for marriage status, and age (see Supporting Information: Appendix A for a discussion of demographic variables). Data from all states except Alaska is used. Dependent variables are either wage and salary annual income or usual hours worked per week, as reported in the ACS. All columns include the interaction terms. Estimations for annual income control for hours worked as this will clearly effect annual income for wage workers. Columns 1 and 3 control for state-level fixed effects, while columns 2 and 4 control for PUMA-level fixed effects. Columns 1 through 4 use a state-level linear time trend, while columns 5 and 6 use a PUMA-level linear time trend. The number of observations may fluctuate slightly due to the use of different control variables but this fluctuation is typically less than 0.5% of the sample All regressions are weighted using survey sampling weights and standard errors are clustered at the state level.

Abbreviation: ACS, American Community Survey; C. PUMA, consistent PUMA; PUMAs, Public Use Microdata Areas.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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our estimation may be hindered by simultaneity between AEWR and employment. While better identification of AEWR would be an interesting topic for future research, minimum wage increases do not seem to have a large influence on guest worker demand.

Table 6 presents the results of our primary specification [Equation (2)], with income and hours as the outcome. On average, the minimum wage does not have a statistically significant relationship with either income or hours. This is consistent with previous findings of small net employment impacts, as

TABLE 7 Impact of minimum wage on migration

	(1) Migrate = 1	(2) Migrate = 1	(3) Migrate inside United States = 1	(4) Migrate inside United States = 1
Minimum wage	0.00252	0.00206	0.000154	
	(0.00346)	(0.00356)	(0.00191)	
Min. wage × noncitizen	−0.00997***	−0.00982***	−0.00115	
	(0.00213)	(0.00223)	(0.00131)	
Min. wage × naturalized	−0.0000973	−0.000672	−0.0000812	
	(0.00263)	(0.00264)	(0.00187)	
Not a US citizen	0.121***	0.119***	0.0113	0.00171
	(0.0202)	(0.0211)	(0.0119)	(0.00221)
Naturalized US citizen	0.00799	0.0140	−0.00167	−0.00230
	(0.0267)	(0.0261)	(0.0164)	(0.00282)
State unemployment rate	−0.00338*	−0.00374*	0.00180	0.000625
	(0.00164)	(0.00175)	(0.00110)	(0.00445)
Min. wage-previous state				−0.00486
				(0.0171)
Observations	112,973	112,944	112,944	11,2914
Year FE	Yes	Yes	Yes	Yes
State FE	Yes	No	No	No
C. PUMA FE	No	Yes	Yes	Yes
State time trend	Yes	Yes	Yes	Yes
C. PUMA time trend	No	No	No	No
Demographic controls	Yes	Yes	Yes	Yes

Note: This sample includes non-manager agricultural workers surveyed in the ACS between 2006 and 2018. Demographic controls include dummies for sex, Hispanic ethnicity, 9 indicators for race, indicators for marriage status and age (See Appendix A for a discussion on demographic variables). Data from all states except Alaska is used. Columns 1 and 2 present the main specifications which are versions of equation 1 but with the outcome variable a binary indicator for having moved in the past year or having moved within the US in the past year and different fixed effects and linear trends. Column 3 uses all internal US moves as the dependent variable, Columns 1 and 2 use contemporaneous, current state minimum wage as the treatment variable while column 4 uses the minimum wage in the previous year in the state the individual moved from (available in ACS data) Ruggles et al. (2020). The number of observations may fluctuate slightly due to the use of different control variables but this fluctuation is typically less than 0.5% of the sample All regressions are weighted using survey sampling weights and standard errors are clustered at the state level.

Abbreviation: C. PUMA, consistent Public Use Microdata Area

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

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well as a large share of farm workers earning more than minimum wage. We do find evidence in some specifications that farm workers who are not US citizens experience a larger increase in income when the minimum wage increases. There is some evidence that the opposite is true for naturalized citizens, though both these findings do not hold in some specifications. Noncitizens earn less than US-born citizens, while naturalized citizens earn more in general. The larger wage effect for noncitizens may reflect that these workers are more likely to earn at or near the minimum wage.[P] Further, if immigrants generally lack bargaining power with employers, their wages may increase relatively more when the minimum wage increases, particularly if there is some wage compression for those already earning above the minimum wage. One caveat for the analysis of annual earnings is that earnings may include wages from nonagricultural jobs. We expect that such wages would not be substantially higher than the agricultural wages of the workers in our sample who classify themselves as agricultural workers. Regardless, an alternative interpretation of our results is that workers who consider themselves primarily employed in agricultural on average do not see an increase in annual earnings from any type of work following an increase in minimum wage.

Table 7 presents the results of minimum wage on migration. The first two columns present the impact of the minimum wage on the likelihood that someone indicates they moved to the state in the last year from either out of state or out of the United States. Columns 3 and 4 present the same results but only for the likelihood that someone moved from another US state. We find that minimum wage does not have a statistically significant relationship with net migration under any specification. However, noncitizens (who are generally more mobile, both within the United States, Basso & Peri, 2020; Schundeln, 2014, and by definition, from abroad) are less likely to have moved into a state with higher minimum wage in the last year. This is in line with results that suggest immigrants may be more likely to move to low minimum wage areas with higher likelihood of employment (Cadena, 2014). The effect disappears when we consider only internal moves (columns 3 to 4), so the result likely reflects the number of recent international moves, which mostly only immigrants make. This result may explain how individual noncitizen farm workers are more likely to remain employed as farm workers when minimum wage increases, as the number of international migrants decreases, which is the most likely group to compete with noncitizens. Column 4 uses the minimum wage in the previous state as the independent variable of interest for internal moves. We find no evidence that raising the minimum wage pushes people out of states or attracts new workers form elsewhere.

5 | CONCLUSION

The short-term impacts of minimum wage increases since 2006 on agricultural employment, including the use of the H-2A program, have been small to negligible. We find that individual farmworkers may experience a one percentage point decrease in the likelihood of employment with a $1 increase in the minimum wage, with a larger effect for native and naturalized citizens. We do not find evidence supporting a relationship between minimum wage changes and the total number of farmworkers or guest workers. However, we do find evidence of fewer international migrants moving to high minimum wage states, consistent with these migrants coming to the United States for employment opportunities and being more vulnerable to economic shocks. This finding may also explain to some degree the different findings between our individual and PUMA specifications; noncitizen farm workers may be more likely to hold positions that international migrants would also take. A lower number of international migrants might also, to some degree, offset any decreases in the employment of domestic farm workers. Our results are largely consistent with the broader minimum wage impacts literature, that is, Cengiz et al. (2019), as well as with minimum wage not being binding under current agricultural labor shortages (Zahniser et al., 2019). Our finding that minimum wage increases typically affect noncitizens

[P] See Borjas and Cassidy (2019); Hall et al. (2010); Pfeffer and Parra (2009) for analysis of this wage gap.

differently than U.S. citizens is similar to results in some other sectors (Orrenius & Zavodny, 2011). These results are robust to different ways for measuring key variables and accounting for local labor market trends, as well as using a synthetic difference-in-differences specification.

Our findings focus on the short-term impacts of minimum wage, which allows us to consider the impacts of changing wages in the context of agriculture's diverse workforce. However, impacts of minimum wage may occur over the long term (Kandilov & Kandilov, 2020; Meer & West, 2016). While discussion of the merits of short versus long-differences methodologies is beyond the scope of this study, we consider our analysis complementary to studies considering long-term impacts. Further, we cannot perfectly account for employers' expectations for future minimum wage increases, after the initial minimum wage changes in our sample. If employers are adjusting during our survey window not just to the current increase in the minimum wage but to the expected future increases (which are still ongoing after the sample window ends), then the results capture both immediate impacts and expectations. This would bias the results towards finding a larger impact, although our results suggest a small or negligible impact at most. Another concern is that the ACS data may not be representative of the farm worker population. We chose the ACS for its information on immigration status, open access, and comparability to other minimum wage studies, for example, Card (1992). While the total number of farmworkers is similar with ACS and the Census of Agriculture, the composition of farmworkers in ACS is an open question. Farmworker data suggests that immigrants make up 70%–80% of domestic farm workers (USDA Economic Research Service, 2020), but they are less than 45% of our main domestic sample. This is consistent with other estimates of immigrants in the agricultural sector though (Desilver, 2017). If there are systematic differences between farm workers captured in the ACS and those that are not, our results may be affected.

There are many still-unanswered questions regarding how rising wages affect the farm workforce. Other datasets, including the Quarterly Census of Employment and National Agricultural Worker Survey, which have been used in farm labor research, for example, Charlton et al. (2022); Richards (2018), may yield additional insights on broader minimum wage impacts, especially if they capture additional or different subpopulations of farm workers. Replicating Cengiz et al. (2019) in a data set that captures hourly wages for farm workers would also improve understanding of farm labor market dynamics. Future research would benefit from the development and application of empirical methodologies that improve the identification of short- and long-term impacts of minimum wage. While our work focuses on minimum wage, agricultural labor research generally should account for workforce heterogeneity. Likewise, future research could analyze how differences in productivity, bargaining power, and economic vulnerability among farm workers influence farm labor market outcomes. Future work could also investigate the impacts of an increase in agricultural guest workers, as the influx of temporary employees could have various effects on local economies, such as housing availability. Methods and datasets used for the analysis of minimum wage impacts may be useful for analyzing other changes to labor regulation, such as overtime and unionization.

Despite substantial increases in several states, our analysis suggests that minimum wage may be less relevant to structural change in agriculture than other market forces and labor regulations. For example, many producers and farm organizations are adjusting to changing overtime regulations in major fruit and vegetable production states (Bohnert, 2021; Clayton, 2021; Vielkind, 2020), where a largely migrant workforce puts in long hours during the time-sensitive harvest season (Beatty et al., 2020). Our analysis shows a focus on net or average impacts in farm labor market research could mask substantial treatment heterogeneity, which has important implications for both policy and research design. Differences in workers' bargaining power and vulnerability to shocks may have a large influence on policy and labor market outcomes. More broadly, our analysis supports claims that widespread labor shortages will drive continued agricultural mechanization (Charlton et al., 2019a), including use of robotics. Further, many agricultural producers face national or international prices, thus are primarily able to adapt to higher labor costs through increasing efficiency, including mechanization.

While farm labor research, especially regarding minimum wage, is limited outside of the US setting, Charlton et al. (2021) document that farm labor shortages are an ongoing issue across a diverse set of

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counties. They also document that the use of migrant labor, including guest workers, is a common feature of agriculture worldwide. Further, minimum wage and other labor regulations are not unique to the United States. Our findings thus are relevant beyond the US setting. Analyses of the economic impact of labor regulations in agriculture needs to account for farm worker heterogeneity, especially when this heterogeneity affects bargaining power or labor market outcomes.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in (1) IPUMS: Steven Ruggles, Sarah Flood, Ronald Goeken, Megan Schouweiler and Matthew Sobek. IPUMS USA: Version 12.0 [data set]. Minneapolis, MN: IPUMS, 2022. https://doi.org/10.18128/D010.V12.0 and (2) Department of Labor Foreign Labor Certification Performance Data at https://www.dol.gov/agencies/eta/foreign-labor/performance and (3) U.S. Bureau of Labor Statistics data reported by FRED at https://fred.stlouisfed.org/. Directions on how to download the data and code used in this study are provided the supplemental online materials.

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Additional supporting information can be found online in the Supporting Information section at the end of this article.

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